Power BI for Developers

2-Week Trainer Guide | 60 Hours

Audience: Frontend/Backend/SQL developers
Delivery model: 2 hours theory + 4 hours lab daily
Goal: SQL/full-stack developer → customer-ready Power BI prototype developer

Trainer Operating Principles

• Teach by business question, not by chart menu.

• Keep demonstrations short; learners reproduce immediately.

• Every KPI must have a definition, source, aggregation and validation method.

• Use SQL as the reconciliation baseline where appropriate.

• Treat Claude as a copilot: draft → inspect → test → validate.

• Separate prototype speed from production hardening.

Day 1 — BI Fundamentals + Power BI Architecture

Learning Objectives

• BI vs OLTP/OLAP

• Facts, dimensions, measures, KPIs

• Power BI Desktop architecture

• Business question → visual

2-Hour Theory / Guided Instruction

• 0:00–0:20: BI vs OLTP/OLAP

• 0:20–0:45: Facts, dimensions, measures, KPIs

• 0:45–1:15: Power BI Desktop architecture

• 1:15–1:45: Business question → visual

• 1:45–2:00: Trainer demonstration, questions and assessment

4-Hour Hands-On Lab

• Hour 1: Retail Sales dataset

• Hour 2: Create KPI cards

• Hour 3: Create trend/region/product visuals

• Hour 4: Add slicers

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

Five business questions → KPI → visual mapping + first report page.

Assessment Answer Guide

• Full credit: business-oriented question, correct KPI, appropriate visual and clear rationale.

Day 2 — Data Connectivity + Power Query

Learning Objectives

• SQL/CSV/Excel connectivity

• Import vs DirectQuery

• Power Query transformations

• Data quality

2-Hour Theory / Guided Instruction

• 0:00–0:20: SQL/CSV/Excel connectivity

• 0:20–0:45: Import vs DirectQuery

• 0:45–1:15: Power Query transformations

• 1:15–1:45: Data quality

• 1:45–2:00: Trainer demonstration, questions and assessment

4-Hour Hands-On Lab

• Hour 1: Messy retail extract

• Hour 2: Fix types/nulls

• Hour 3: Merge customer master

• Hour 4: Append yearly sales

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

Data-quality report documenting issue, impact, transformation and validation.

Assessment Answer Guide

• Full credit: issue → business impact → transformation → validation, with no unexplained data changes.

Day 3 — Semantic Modelling + Star Schema

Learning Objectives

• Fact/dimension design

• Relationships/cardinality

• Date dimension

• Filter direction

• Modelling anti-patterns

2-Hour Theory / Guided Instruction

• 0:00–0:20: Fact/dimension design

• 0:20–0:45: Relationships/cardinality

• 0:45–1:15: Date dimension

• 1:15–1:45: Filter direction

• 1:45–2:00: Modelling anti-patterns

4-Hour Hands-On Lab

• Hour 1: Build Retail star schema

• Hour 2: Create date table

• Hour 3: Create hierarchy

• Hour 4: Test filter propagation

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

Star-schema model diagram + relationship rationale.

Assessment Answer Guide

• Expected: dimensions on the one-side, FactSales on the many-side, clean star schema, dedicated date dimension.

Day 4 — Visualization + Report UX

Learning Objectives

• Visual selection

• Layout and hierarchy

• Slicers/tooltips

• Conditional formatting

• Accessibility

2-Hour Theory / Guided Instruction

• 0:00–0:20: Visual selection

• 0:20–0:45: Layout and hierarchy

• 0:45–1:15: Slicers/tooltips

• 1:15–1:45: Conditional formatting

• 1:45–2:00: Accessibility

4-Hour Hands-On Lab

• Hour 1: Build Executive page

• Hour 2: Build Analysis page

• Hour 3: Add tooltip

• Hour 4: Redesign cluttered page

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

Two-page report + 10-point UX review.

Assessment Answer Guide

• Full credit: hierarchy, consistent formatting, useful slicers, minimal clutter, readable labels and actionable comparisons.

Day 5 — DAX Essentials

Learning Objectives

• Measures vs calculated columns

• SUM/COUNT/DISTINCTCOUNT

• DIVIDE

• CALCULATE/FILTER

• Time intelligence

2-Hour Theory / Guided Instruction

• 0:00–0:20: Measures vs calculated columns

• 0:20–0:45: SUM/COUNT/DISTINCTCOUNT

• 0:45–1:15: DIVIDE

• 1:15–1:45: CALCULATE/FILTER

• 1:45–2:00: Time intelligence

4-Hour Hands-On Lab

• Hour 1: Build 10 KPI measures

• Hour 2: YTD/LY/YoY

• Hour 3: Test filter context

• Hour 4: Reconcile to SQL

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

10-measure KPI library + five plain-English DAX explanations.

Assessment Answer Guide

• Expected measures include Revenue, Cost, Profit, Margin, Orders, Customers, AOV, YTD, LY and YoY. Explanations must reflect filter context.

Day 6 — Rapid Customer Prototyping

Learning Objectives

• Requirement decomposition

• KPI dictionary

• Wireframing

• Prototype vs production

2-Hour Theory / Guided Instruction

• 0:00–0:20: Requirement decomposition

• 0:20–0:45: KPI dictionary

• 0:45–1:15: Wireframing

• 1:15–1:45: Prototype vs production

• 1:45–2:00: Trainer demonstration, questions and assessment

4-Hour Hands-On Lab

• Hour 1: Healthcare requirement

• Hour 2: Model + measures

• Hour 3: 3-page prototype

• Hour 4: 5-minute customer demo

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

Requirement-to-KPI-to-visual traceability matrix + three-page prototype.

Assessment Answer Guide

• Full credit: requirement is decomposed before building; KPI definitions and assumptions are explicit; prototype demonstrates stated questions.

Day 7 — Interactive Report Development

Learning Objectives

• Drill-down/drill-through

• Bookmarks/buttons

• Navigation

• Dynamic titles

• Tooltips

2-Hour Theory / Guided Instruction

• 0:00–0:20: Drill-down/drill-through

• 0:20–0:45: Bookmarks/buttons

• 0:45–1:15: Navigation

• 1:15–1:45: Dynamic titles

• 1:45–2:00: Tooltips

4-Hour Hands-On Lab

• Hour 1: Build drill paths

• Hour 2: Add navigation

• Hour 3: Provider/customer detail

• Hour 4: 10 journey tests

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

10-step interaction test script + interactive report.

Assessment Answer Guide

• Full credit: drill-down and drill-through are correctly distinguished and user journeys are tested under multiple filter states.

Day 8 — Claude-Assisted BI Development

Learning Objectives

• AI-assisted workflow

• DAX prompting

• Power Query/SQL prompting

• Requirement analysis

• AI validation/governance

2-Hour Theory / Guided Instruction

• 0:00–0:20: AI-assisted workflow

• 0:20–0:45: DAX prompting

• 0:45–1:15: Power Query/SQL prompting

• 1:15–1:45: Requirement analysis

• 1:45–2:00: AI validation/governance

4-Hour Hands-On Lab

• Hour 1: Generate DAX

• Hour 2: Generate M code

• Hour 3: Requirement analysis

• Hour 4: UX critique

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

10-prompt Claude library + prompt log + two identified AI errors.

Assessment Answer Guide

• Full credit: prompts constrain schema, require explanations, and include validation. Learner identifies at least two AI errors/assumptions.

Day 9 — MicroStrategy/Cognos Legacy Migration

Learning Objectives

• Legacy inventory

• Metric/attribute mapping

• Prompt/filter mapping

• Reconciliation

• Gap/risk management

2-Hour Theory / Guided Instruction

• 0:00–0:20: Legacy inventory

• 0:20–0:45: Metric/attribute mapping

• 0:45–1:15: Prompt/filter mapping

• 1:15–1:45: Reconciliation

• 1:45–2:00: Gap/risk management

4-Hour Hands-On Lab

• Hour 1: Analyze legacy report

• Hour 2: Build mapping matrix

• Hour 3: Recreate model/DAX

• Hour 4: Validate outputs

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

Legacy migration matrix + gap/risk log + migrated page.

Assessment Answer Guide

• Expected mapping: metric→measure, prompt→slicer/filter, attribute→dimension, drill→drill-down/through. Numerical reconciliation is mandatory.

Day 10 — Capstone + Certification

Learning Objectives

• Customer delivery

• Technical validation

• Presentation

• Production hardening

2-Hour Theory / Guided Instruction

• 0:00–0:20: Customer delivery

• 0:20–0:45: Technical validation

• 0:45–1:15: Presentation

• 1:15–1:45: Production hardening

• 1:45–2:00: Trainer demonstration, questions and assessment

4-Hour Hands-On Lab

• Hour 1: Build/finalize capstone

• Hour 2: Validation sheet

• Hour 3: 10-minute demo

• Hour 4: Q&A;

Trainer Checkpoint

Require learners to show the artifact, explain one design decision, and validate at least one result.

Daily Assignment & Deliverable

Capstone package + validation sheet + customer demo.

Assessment Answer Guide

• Certification requires 70%+ and no critical failure in KPI correctness, model integrity or capstone validation.

Assessment & Certification Rubric

Area	Weight
BI fundamentals	10%
Power Query / data preparation	15%
Semantic modelling	15%
DAX & KPI correctness	15%
Visualization & UX	10%
Interactive development	10%
Claude-assisted productivity	10%
Legacy migration	5%
Capstone delivery	10%

Pass: 70% overall + no critical failure. **90–100:** Advanced Practitioner. **80–89:** Practitioner. **70–79:** Certified Foundation. **<70:** Reassessment.

Critical Failures

• Materially incorrect KPI results without detection.

• Fundamentally broken relationships/model.

• Unable to explain own DAX or report logic.

• No reconciliation of important numbers.

• Presenting unvalidated AI output as correct.